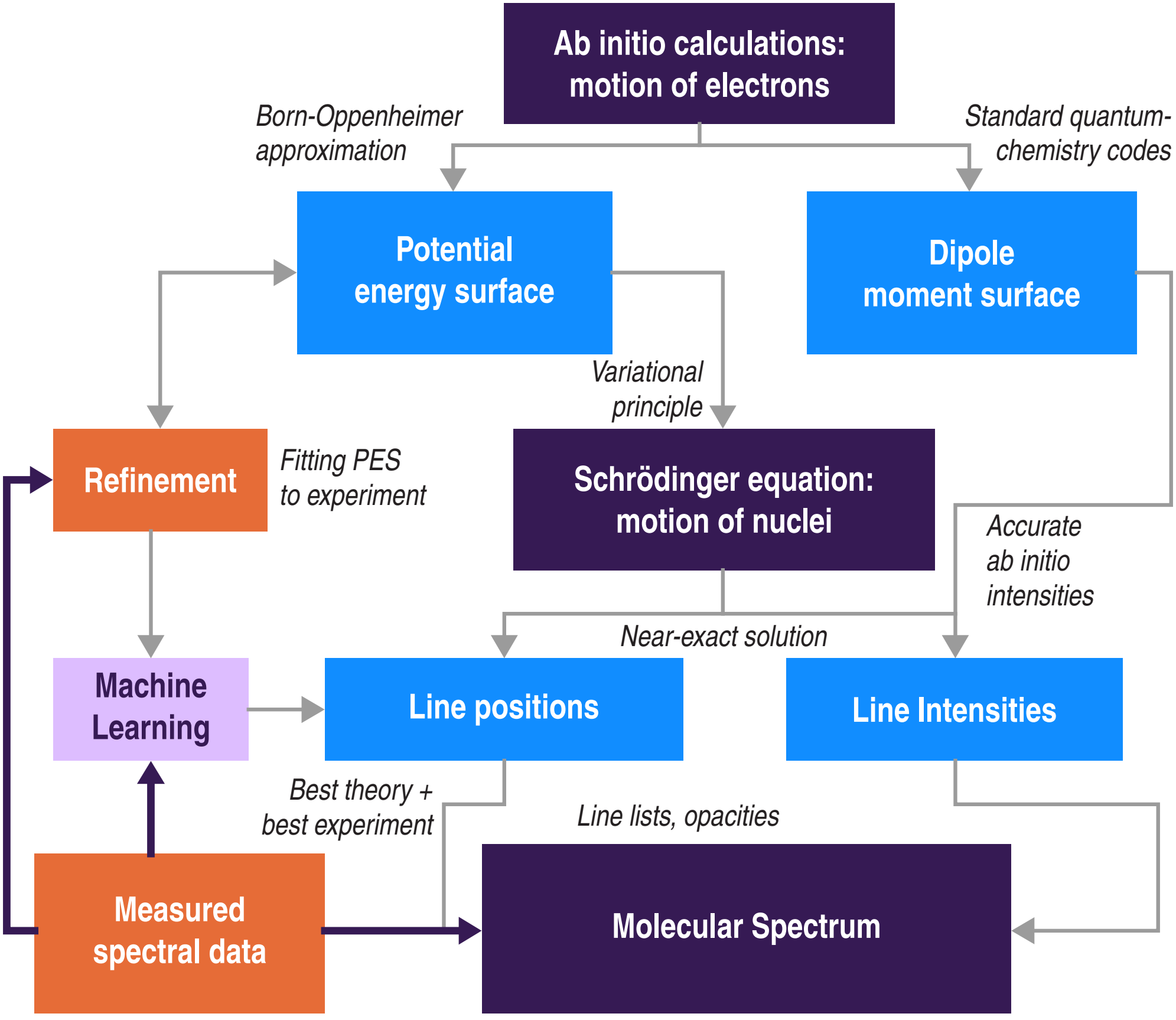
Ab initio calculations: motion of electrons
Born-Oppenheimer approximation
Standard quantum-chemistry codes
Potential energy surface
Dipole moment surface
Variational principle
Refinement
Fitting PES to experiment
Schrödinger equation: motion of nuclei
Accurate ab initio intensities
Near-exact solution
Machine Learning
Line positions
Line Intensities
Best theory + best experiment
Line lists, opacities
Measured spectral data
Molecular Spectrum